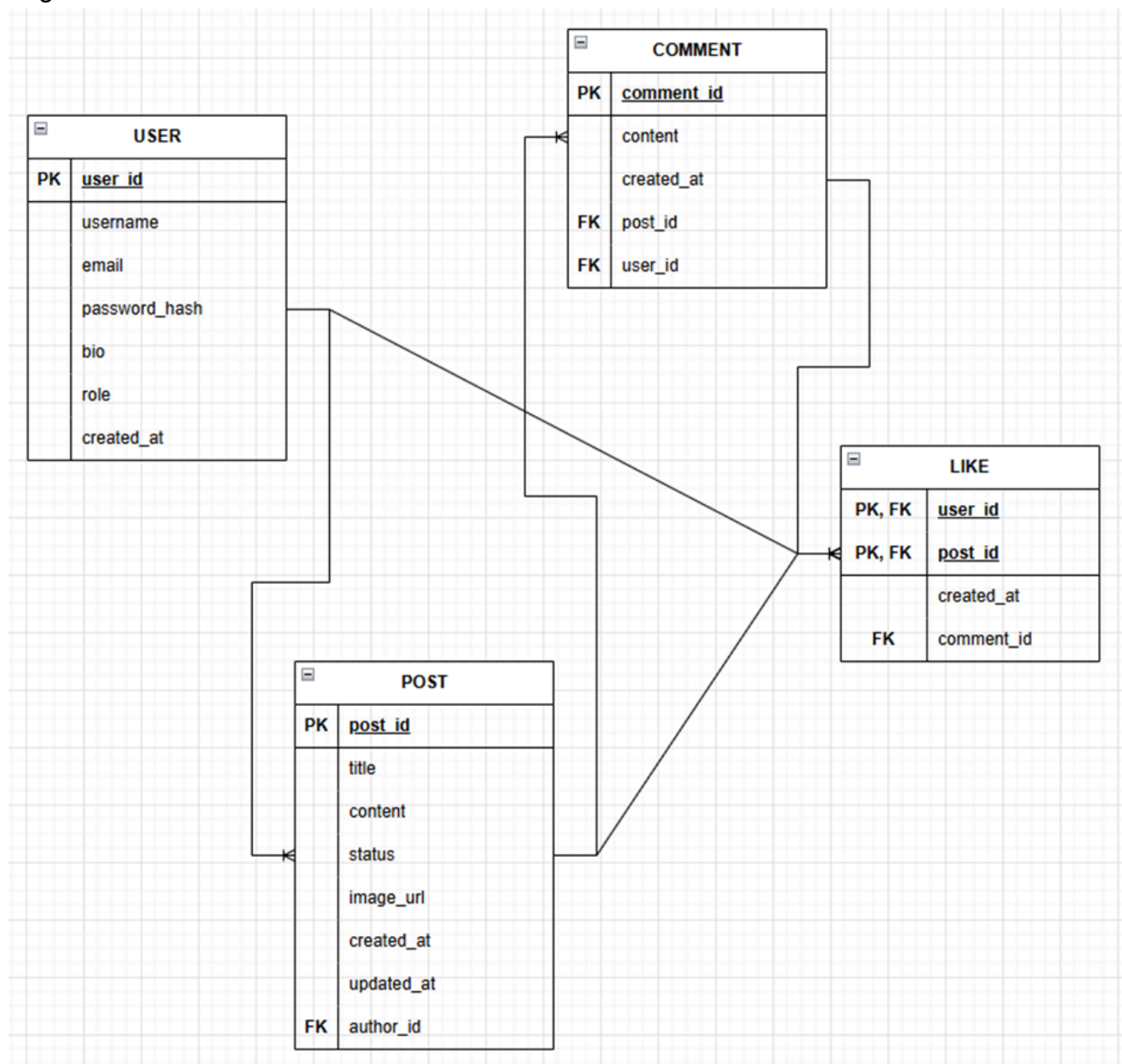
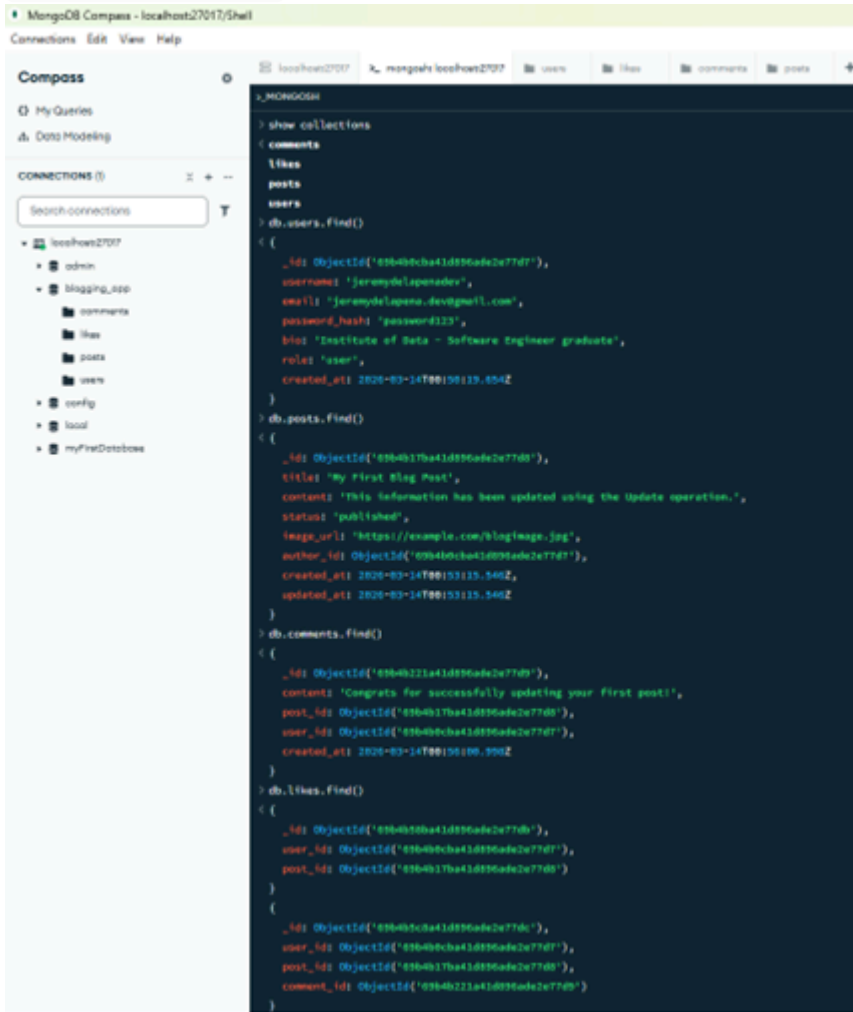


Module 8 - Lab Exercise 2 (via MongoDB)

Logical Model Used:



show collections



CRUD Operations

`db.users.insertOne()`

`db.users.find()`

Compass

{ } My Queries

⚙ Data Modeling

CONNECTIONS (1)

- localhost:27017
 - admin
 - config
 - local
 - myFirstDatabase

localhost:27017

mongosh: localhost:27017



> _MONGOSH

> show databases

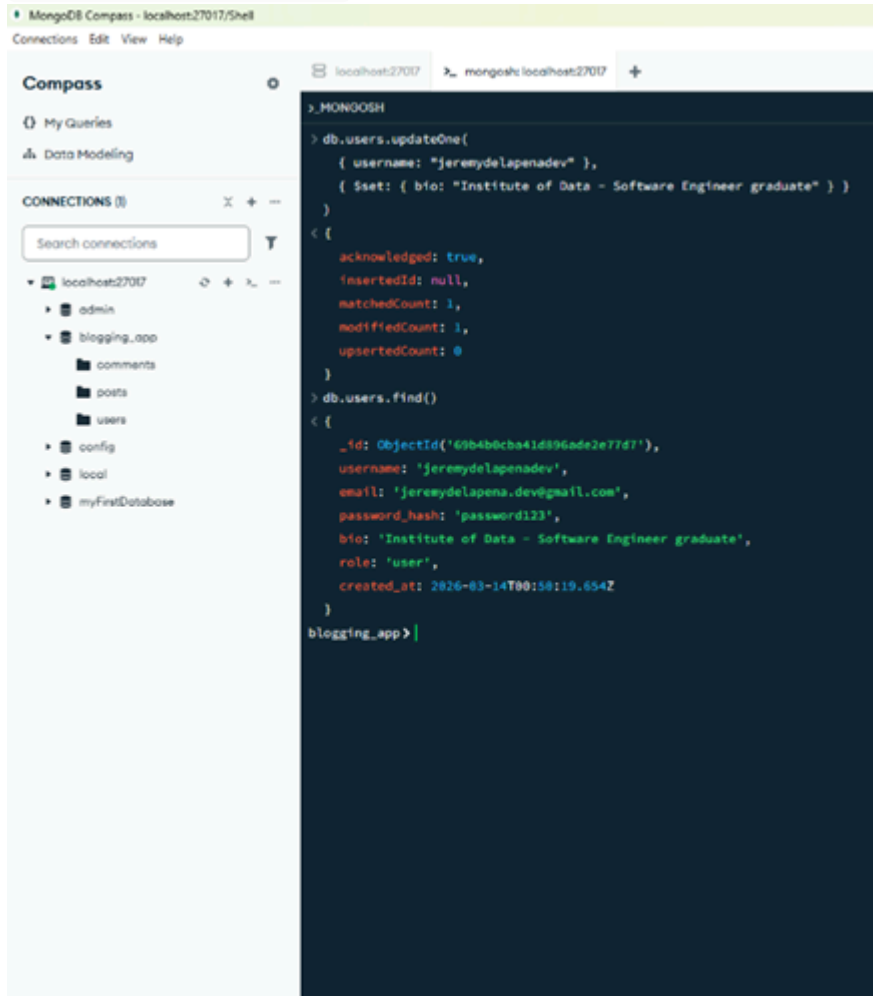
```
< admin          40.00 KiB
  config         108.00 KiB
  local           76.00 KiB
  myFirstDatabase 12.00 KiB
```

> use blogging_app

< switched to db blogging_app

```
> db.users.insertOne({
  username: "jeremydelapenadev",
  email: "jeremydelapena.dev@gmail.com",
  password_hash: "password123",
  bio: "Software Engineering student",
  role: "user",
  created_at: new Date()
})
< {
  acknowledged: true,
  insertedId: ObjectId('69b4b0cba41d896ade2e77d7')
}
> db.users.find()
< {
  _id: ObjectId('69b4b0cba41d896ade2e77d7'),
  username: 'jeremydelapenadev',
  email: 'jeremydelapena.dev@gmail.com',
  password_hash: 'password123',
  bio: 'Software Engineering student',
  role: 'user',
  created_at: 2026-03-14T00:50:19.654Z
}
blogging_app> |
```

db.users.updateOne()



db.posts.insertOne()

db.posts.find()

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

local

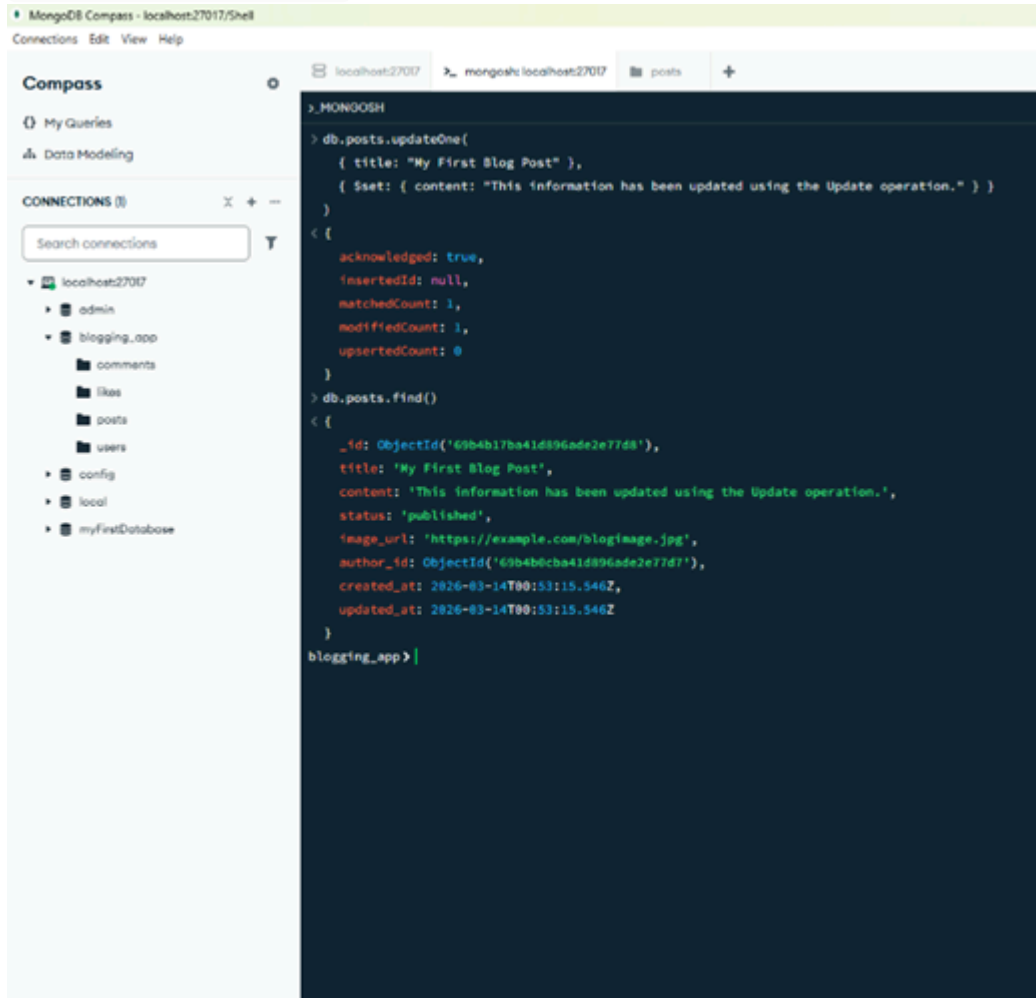
myFirstDatabase

localhost:27017

mongo: mongo localhost:27017

```
> use blogging_app
switched to db blogging_app
> db.users.insertOne({
  username: "jeremydelapenadev",
  email: "jeremydelapena.dev@gmail.com",
  password_hash: "password123",
  bio: "Software Engineering student",
  role: "user",
  created_at: new Date()
})
< {
  acknowledged: true,
  insertedId: ObjectId('69b4b8c8a41d896ade2e77d7')
}
> db.users.find()
< {
  _id: ObjectId('69b4b8c8a41d896ade2e77d7'),
  username: 'jeremydelapenadev',
  email: 'jeremydelapena.dev@gmail.com',
  password_hash: 'password123',
  bio: 'Software Engineering student',
  role: 'user',
  created_at: 2026-03-14T00:58:19.654Z
}
> db.posts.insertOne({
  title: "My First Blog Post",
  content: "The information for the post is here.",
  status: "published",
  image_url: "https://example.com/blogimage.jpg",
  author_id: ObjectId("69b4b8c8a41d896ade2e77d7"),
  created_at: new Date(),
  updated_at: new Date()
})
< {
  acknowledged: true,
  insertedId: ObjectId('69b4b17ba41d896ade2e77d8')
}
blogging_app>
```

```
db.posts.updateOne()
```



The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel lists the local database connection. The main panel displays the 'posts' collection. The command window shows the following execution:

```
> db.posts.updateOne(
  { title: "My First Blog Post" },
  { $set: { content: "This information has been updated using the Update operation." } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.posts.find()
< {
  _id: ObjectId('69b4b17ba41d896ade2e77d8'),
  title: 'My First Blog Post',
  content: 'This information has been updated using the Update operation.',
  status: 'published',
  image_url: 'https://example.com/blogimage.jpg',
  author_id: ObjectId('69b4b0c8a41d896ade2e77d1'),
  created_at: 2026-03-14T00:53:15.546Z,
  updated_at: 2026-03-14T00:53:15.546Z
}
blogging_app>
```

```
db.comments.insertOne()
```

```
db.comments.find()
```

db.comments.deleteOne()

```
MongoDB Compass - localhost:27017/Shell
Connections Edit View Help

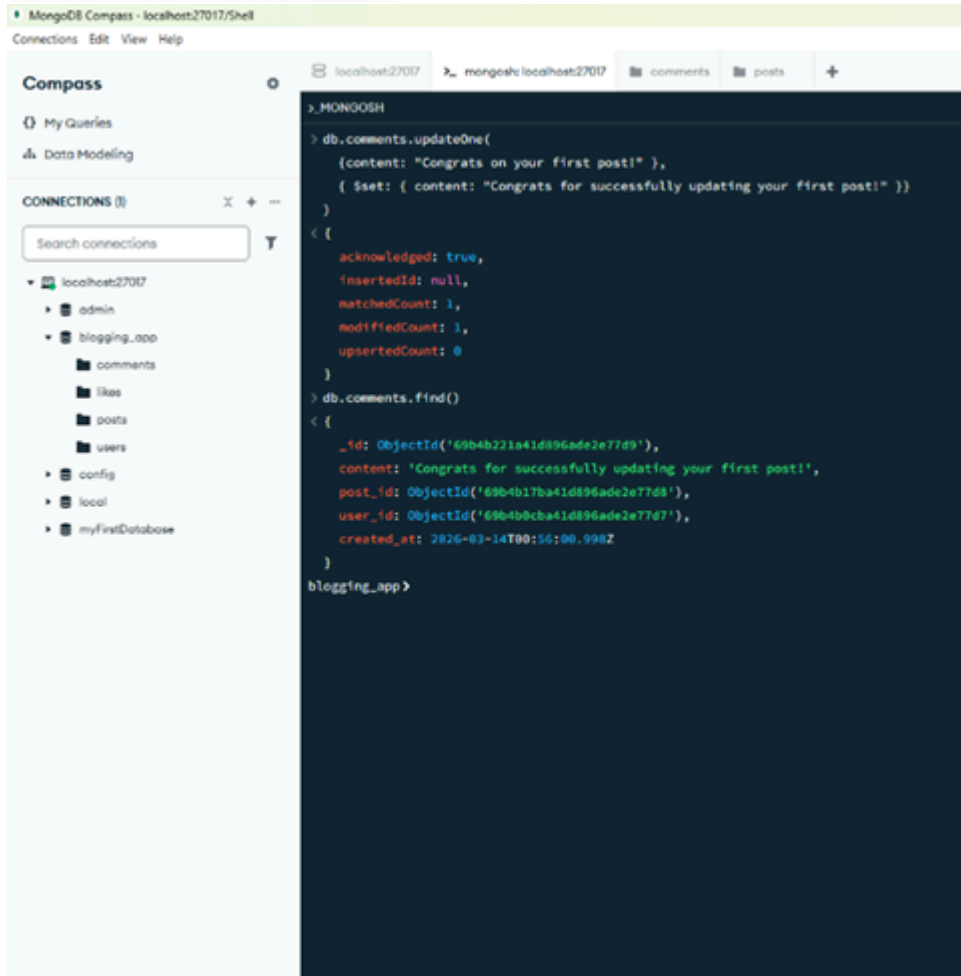
Compass
My Queries
Data Modeling

CONNECTIONS [1]
Search connections

localhost27017
  admin
  blogging_app
    comments
    likes
    posts
    users
  config
  local
  myFirstDatabase

>_MONGODB_
> db.comments.insertOne({
  content: "Comment#2",
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  user_id: ObjectId("69b4b0c8a41d896ade2e77d7"),
  created_at: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId('69b4b45fa41d896ade2e77dd')
}
> db.comments.find()
{
  _id: ObjectId('69b4b221a41d896ade2e77d9'),
  content: 'Congrats for successfully updating your first post!',
  post_id: ObjectId('69b4b17ba41d896ade2e77d8'),
  user_id: ObjectId('69b4b0c8a41d896ade2e77d7'),
  created_at: 2026-03-14T00:54:00.998Z
}
{
  _id: ObjectId('69b4b45fa41d896ade2e77dd'),
  content: 'Comment#2',
  post_id: ObjectId('69b4b17ba41d896ade2e77d8'),
  user_id: ObjectId('69b4b0c8a41d896ade2e77d7'),
  created_at: 2026-03-14T00:14:07.887Z
}
> db.comments.deleteOne(
  { content: 'Comment#2' }
)
{
  acknowledged: true,
  deletedCount: 1
}
> db.comments.find()
{
  _id: ObjectId('69b4b221a41d896ade2e77d9'),
  content: 'Congrats for successfully updating your first post!',
  post_id: ObjectId('69b4b17ba41d896ade2e77d8'),
  user_id: ObjectId('69b4b0c8a41d896ade2e77d7'),
  created_at: 2026-03-14T00:54:00.998Z
}
blogging_app>
```

```
db.comments.updateOne()
```



The screenshot shows the MongoDB Compass application. On the left, the 'CONNECTIONS (1)' panel lists the connected database 'myFirstDatabase'. The main area displays the 'comments' collection. The command window on the right shows the following execution:

```
> db.comments.updateOne(
  {content: "Congrats on your first post!" },
  { $set: { content: "Congrats for successfully updating your first post!" }}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.comments.find()
< {
  _id: ObjectId('69b4b221a41d896ade2e77d9'),
  content: 'Congrats for successfully updating your first post!',
  post_id: ObjectId('69b4b17ba41d896ade2e77d8'),
  user_id: ObjectId('69b4b9c8a41d896ade2e77d7'),
  created_at: 2026-03-14T00:56:00.998Z
}
blogging_app>
```

```
db.likes.insertOne()
```

```
db.likes.find()
```


Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

blogging_app

comments

posts

users

config

local

myFirstDatabase

localhost:27017

mongosh localhost:27017

```
> use blogging_app
switched to db blogging_app

> db.comments.insertOne({
  content: "Congrats on your first post!",
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  user_id: ObjectId("69b4b0cba41d896ade2e77d7"),
  created_at: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId("69b4b221a41d896ade2e77d9")
}

> db.comments.find()
{
  _id: ObjectId("69b4b221a41d896ade2e77d9"),
  content: "Congrats on your first post!",
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  user_id: ObjectId("69b4b0cba41d896ade2e77d7"),
  created_at: 2026-03-14T00:56:00.998Z
}

> db.likes.insertOne({
  user_id: ObjectId("69b4b0cba41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  comment_id: ObjectId("69b4b221a41d896ade2e77d9"),
  created_at: new Date()
})
{
  acknowledged: true,
  insertedId: ObjectId("69b4b297a41d896ade2e77da")
}

> db.likes.find()
{
  _id: ObjectId("69b4b297a41d896ade2e77da"),
  user_id: ObjectId("69b4b0cba41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  comment_id: ObjectId("69b4b221a41d896ade2e77d9"),
  created_at: 2026-03-14T00:57:59.679Z
}

blogging_app>
```

db.likes.deleteOne()

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (0)

Search connections

- localhost:27017
 - admin
 - blogging_app
 - comments
 - likes
 - posts
 - users
 - config
 - local
 - myFirstDatabase

localhost:27017 mongosh localhost:27017 likes comments posts +

```
> _MONGOOSH
> db.likes.deleteOne(
  {post_id: ObjectId("69b4b17ba41d896ade2e77d8")}
)
< {
  acknowledged: true,
  deletedCount: 1
}
> db.likes.find()
<
> db.likes.insertOne({
  user_id: ObjectId("69b4b8c8a41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8")
})
< {
  acknowledged: true,
  insertedId: ObjectId("69b4b58ba41d896ade2e77db")
}
> db.likes.find()
< {
  _id: ObjectId("69b4b58ba41d896ade2e77db"),
  user_id: ObjectId("69b4b8c8a41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8")
}
> db.likes.insertOne({
  user_id: ObjectId("69b4b8c8a41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8"),
  comment_id: ObjectId("69b4b221a41d896ade2e77d9")
})
< {
  acknowledged: true,
  insertedId: ObjectId("69b4b5c8a41d896ade2e77dc")
}
> db.likes.find()
< {
  _id: ObjectId("69b4b58ba41d896ade2e77db"),
  user_id: ObjectId("69b4b8c8a41d896ade2e77d7"),
  post_id: ObjectId("69b4b17ba41d896ade2e77d8")
}
```